$z = a + bi$ $\quad 7 + 4i \rightarrow$ Rectangular

$z = r \cdot \text{cis}(\theta)$, $r = \sqrt{a^2+b^2}$, $\theta = \tan^{-1}\left(\frac{b}{a}\right) \rightarrow$ Polar

$r \cdot e^{i\theta} \rightarrow$ Exponencial

$7+4i \rightarrow a = 7 \quad r = \sqrt{7^2+4^2} = \sqrt{65}$

$b = 4 \quad \theta = \tan^{-1}\left(\frac{4}{7}\right) = 0,51$

$$z = \sqrt{65} \cdot \text{cis}(0,51)$$

$$\text{cis}(\theta) = \cos(\theta) + i\,\text{sen}(\theta)$$

$z = a + bi$

$\theta = \tan\left(\frac{b}{a}\right)$

Cuadrante	a	b	Que hago?
I	+	+	NADA
II	−	+	$+\pi \rightarrow \theta = \tan^{-1}\left(\frac{b}{a}\right) + \pi$
III	−	−	$-\pi \rightarrow \theta = \tan^{-1}\left(\frac{b}{a}\right) - \pi$
IV	+	−	NADA

$a = 0$ y $b > 0 \quad \theta = \frac{\pi}{2} \quad 3i$

$a = 0$ y $b < 0 \quad \theta = -\frac{\pi}{2} \quad -3i$

Teorema de demoivre

$r\,cis(\theta)$ $e^{i\theta}$

Z_1, Z_2 Ambos en forma polar o exponencial

$Z_1 \cdot Z_2$

$r_1\,cis(\theta_1) \cdot r_2 \cdot cis(\theta_2)$ } multiplicacion

$r_1 \cdot r_2 \cdot cis(\theta_1 + \theta_2)$

$Z_1 = 3 \cdot cis(7) \qquad Z_2 = 7 \cdot cis(4)$

$3 \cdot 7 \cdot cis(7+4)$

$$\frac{Z_1}{Z_2} = \frac{r_1\,cis(\theta_1)}{r_2 \cdot cis(\theta_2)}$$

$$\frac{r_1}{r_2} \cdot cis(\theta_1 - \theta_2)$$

} Division

$Z_1 = 3 \cdot cis(7) \qquad Z_2 = 7 \cdot cis(4)$

$$\frac{3}{7} \cdot cis(7-4)$$

$[r\,cis(\theta)]^n$, n es un numero

$r^n \cdot cis(\theta \cdot n)$

$[r \cdot e^{i\theta}] = r^n \cdot e^{i(\theta \cdot n)}$

$r_1 \cdot e^{i\theta_1} \cdot r_2 \cdot e^{i\theta_2} = r_1 \cdot r_2 \cdot e^{i(\theta_1+\theta_2)}$ } Exponencial

$$\frac{r_1 \cdot e^{i\theta_1}}{r_2 \cdot e^{i\theta_2}} = \frac{r_1}{r_2} \cdot e^{i(\theta_1-\theta_2)}$$ } Exponencial

1) Pasar todo a polar

2) Recordar $+\pi$ o $-\pi$ o nada

3) Aplicar el exponente $[r\,cis(\theta)]^n = r^n \cdot cis(\theta \cdot n)$

7. **[5 puntos]** Calcule y exprese el número complejo $z = \sqrt{(1+i)^{-i}} + \sqrt{(2\sqrt{3}-2i)^5}$ en la forma rectangular.

z_1 z_2

$z_2 = 2\sqrt{3} - 2i$ $\quad a = 2\sqrt{3}$ $\quad r = \sqrt{a^2+b^2}$

a $\quad b$ $\quad b = 2$ $\quad \theta = \tan^{-1}\left(\frac{b}{a}\right)$

$r = \sqrt{(2\sqrt{3})^2 + (-2)^2} = 4$

$\theta = \tan^{-1}\left(\frac{-2}{2\sqrt{3}}\right) = \frac{-\pi}{6}$

— + → $+\pi$

— — → $-\pi$

Radianes

$z_2^5 = \left[4\,\mathrm{cis}\left(\frac{-\pi}{6}\right)\right]^5$ ley n

r $\quad \theta$

$[r\,\mathrm{cis}(\theta)]^n = r^n \cdot \mathrm{cis}(\theta \cdot n)$

$4^5 \cdot \mathrm{cis}\left(\frac{-\pi}{6} \cdot 5\right)$

$1024 \cdot \mathrm{cis}\left(\frac{-5\pi}{6}\right)$

$z_2 = (1+i)^{-i}$ → No es un numero suelto, usar ln

$\ln(z_2) = \ln(1+i)^{-i}$ $\qquad \ln(x)^y = y\ln(x)$

$= (-i)\ln(1+i)$

No se toca. Esta de adentro si, en exponencial por que esta dentro de un ln → $re^{i(\theta)}$

$z = 1+i$ $\quad a = 1$ $\quad b = 1$

$r = \sqrt{1^2+1^2} = \sqrt{2}$

$\theta = \tan^{-1}\left(\frac{1}{1}\right) = \frac{\pi}{4}$

$z = \sqrt{2} \cdot e^{i\left(\frac{\pi}{4}\right)}$

$z = \sqrt{2} \cdot e^{i\left(\frac{\pi}{4}\right)}$ $\qquad \ln(a \cdot b) = \ln(a) + \ln(b)$

$= (-i)\ln(1+i)$ $\qquad \ln(e)^x = x$

$= (-i)\left[\ln\left(\underset{a}{\sqrt{2}} \cdot \underset{b}{e^{i\left(\frac{\pi}{4}\right)}}\right)\right]$

$(-i) \cdot \left[\ln(\sqrt{2}) + \ln(e)^{i\frac{\pi}{4}}\right]$

$(-i) \cdot \left[\ln(\sqrt{2}) + i\frac{\pi}{4}\right]$ $\qquad -i \cdot i = -i^2 = -1 \cdot -1 = 1$

$-\ln(\sqrt{2})i + \frac{\pi}{4}$ $\qquad \ln(x)i + \theta$

$e^{\frac{\pi}{4}} \cdot \text{cis}(-\ln(\sqrt{2}))$ $\qquad e^{\theta} \cdot \text{cis}(\ln(x))$

la i se va

$\text{cis}(\theta) = \cos(\theta) + i\,\text{sen}(\theta)$

$= (1+i)^{-i} + (2\sqrt{3} - 2i)^5$

$e^{\frac{\pi}{4}} \cdot \text{cis}(-\ln(\sqrt{2})) + 1024 \cdot \text{cis}\left(\frac{-5\pi}{6}\right)$

$e^{\frac{\pi}{4}}\left[\cos(-\ln(\sqrt{2})) + i\,\text{sen}(-\ln(\sqrt{2}))\right] + 1024\left[\cos\left(\frac{-5\pi}{6}\right) + i\,\text{sen}\left(\frac{-5\pi}{6}\right)\right]$

$2,06 - 0,74i - 886,81 - 512,74i$

$\boxed{-884,81 - 512,74i}$

De radianes a grados

radianes. $\frac{180}{\pi}$

Cuadrante	a	b
1) $0 < \theta < \frac{\pi}{2}$ $0 < a < 90$	> 0	> 0
2) $\frac{\pi}{2} < \theta < \pi$ $90 < a < 180$	< 0	> 0
3) $\pi < \theta < \frac{3\pi}{2}$ $180 < a < 270$	< 0	< 0
4) $\frac{3\pi}{2} < \theta < 2\pi$ $270 < a < 360$	> 0	< 0

En cualquier caso que el arg NO sea $\pm\pi$ y $\pm\frac{\pi}{2}$ para sacar el angulo se $\theta = \left(\frac{b}{a} = \tan(\theta)\right)$

5. **[5 puntos]** Determine $z \in \mathbb{C}$ que satisface simultanéamente las siguientes condiciones :

- $|\overline{z} + (1 - i)| = 5$
- $Arg(z - (1 + 2i)) = \frac{3\pi}{4}$

$z = a + bi$

$\overline{z} = a - bi$ $\quad \overline{z} = 2 - 3i$

$z = 2 + 3i$

$|\overline{z} + (1 - i)| = 5$

$|\overline{a+bi} + (1 - i)| = 5$

$|a - bi + 1 - i| = 5$

$|(a+1) + (-b-1)i| = 5$

Todo ^{2} y quita el i

$(a+1)^2 + (-b-1)^2 = 25$

a $\quad$ b

$Arg(z - (1 + 2i)) = \frac{3\pi}{4}$ $\neq \pm\pi \vee \pm\frac{\pi}{2}$

$\frac{b}{a} = \tan(\theta)$

$a + bi - 1 - 2i = \frac{3\pi}{4} \rightarrow \frac{3\pi}{4}, \frac{180}{\pi}$

$= 135°$

$(a-1) + (b-2)i = \frac{3\pi}{4}$

a $\quad$ b $\quad$ $\frac{3\pi}{4}$

$90 < 135 < 180$

$a < 0 \wedge b > 0$

Calcu $= -1$

$$\frac{b-2}{a-1} = \tan\left(\frac{3\pi}{4}\right)$$

$$\frac{b-2}{a-1} = -1$$

$b - 2 = 1 - a$ $\quad \begin{cases} b = -1 - a \\ a = 3 - b \end{cases}$

$(a+1)^2 + (-b-1)^2 = 25 \qquad a = 3-b$

$a \qquad b$

$(3-b+1)^2 + (-b-1)^2 = 25$

$(4-b)^2 + (-b-1)^2 = 25 \qquad (a+b)^2 = a^2+2ab+b^2$

$(-a-b)^2 = a^2+2ab+b^2$

$16-8b+b^2+b^2+2b+1 = 25 \qquad (a-b)^2 = a^2-2ab+b^2$

$16-8b+b^2+b^2+2b+1-25 = 0$

$2b^2-6b+-8 = 0$

$b = -1 \qquad \boxed{b = 4} \qquad a < 0 \wedge b > 0$

$a = 3-b$

$a = 3-4$

$a = -1$

$\boxed{z = -1+4i}$

$$A = \begin{bmatrix} 3 & 4 \\ 1 & 1 \end{bmatrix} \qquad I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Paso 1: Aumentar

$$= \left[\begin{array}{cc|cc} 3 & 4 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{cc|cc} \textcircled{3} & 4 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{array}\right]$$

$\frac{1}{3} \cdot F1$

$$\left[\begin{array}{cc|cc} \textcircled{3} & 4 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{array}\right] \qquad 3x = 1 \quad x = \frac{1}{3}$$

$\frac{1}{3} \cdot F1$

$$\left[\begin{array}{cc|cc} \textcircled{1} & \frac{4}{3} & \frac{1}{3} & 0 \\ 1 & 1 & 0 & 1 \end{array}\right]$$

K. Pivote + x, Fila a Cambiar

$-1 \cdot F1 + F2$

$$\left[\begin{array}{cc|cc} \textcircled{1} & \frac{4}{3} & \frac{1}{3} & 0 \\ 0 & -\frac{1}{3} & -\frac{1}{3} & 1 \end{array}\right]$$

$$\left[\begin{array}{cc|cc} 1 & \frac{4}{3} & \frac{1}{3} & 0 \\ 0 & \textcircled{-\frac{1}{3}} & -\frac{1}{3} & 1 \end{array}\right]$$

$$\left[\begin{array}{cc|cc} 1 & \frac{4}{3} & \frac{1}{3} & 0 \\ 0 & \textcircled{1} & 1 & -3 \end{array}\right] \qquad -\frac{1}{3}x = 1 \quad x = -3$$

$-3 \cdot F2$

$$\xrightarrow{-\frac{4}{3}\cdot F2+F1} \left[\begin{array}{cc|cc} 1 & 0 & -1 & 4 \\ 0 & 1 & 1 & -3 \end{array}\right]$$

$$A^{-1} = \begin{bmatrix} -1 & 4 \\ 1 & -3 \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 7 & 4 \\ 3 & 2 & 3 \end{bmatrix}$$

$F1 \leftrightarrow F2$

$$\begin{bmatrix} 1 & 7 & 4 \\ 0 & 1 & 2 \\ 3 & 2 & 3 \end{bmatrix}$$

$$\left[\begin{array}{ccc|ccc} 1 & 7 & 4 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 & 1 & 0 \\ 3 & 2 & 3 & 0 & 0 & 1 \end{bmatrix}$$

K · Fila pivote + Fila a cambiar

$$\left[\begin{array}{ccc|ccc} \boxed{1} & 7 & 4 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 & 1 & 0 \\ 0 & -22 & -11 & -3 & 0 & 1 \end{bmatrix}$$

$-3 \cdot F1 + \widetilde{F3}$

$$\left[\begin{array}{ccc|ccc} 1 & 7 & 4 & 1 & 0 & 0 \\ 0 & \boxed{1} & 2 & 0 & 1 & 0 \\ 0 & -22 & -11 & -3 & 0 & 1 \end{bmatrix}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -10 & 1 & -7 & 0 \\ 0 & \boxed{1} & 2 & 0 & 1 & 0 \\ 0 & -22 & -11 & -3 & 0 & 1 \end{bmatrix}$$

$-7 \cdot F2 + \widetilde{F1}$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -10 & 1 & -7 & 0 \\ 0 & \boxed{1} & 2 & 0 & 1 & 0 \\ 0 & 0 & 33 & -3 & 22 & 1 \end{bmatrix}$$

$22 \cdot F2 + \widetilde{F3}$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -10 & 1 & -7 & 0 \\ 0 & 1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 33 & -3 & 22 & 1 \end{array}\right]$$

$\frac{1}{33}\cdot F3$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -10 & 1 & -7 & 0 \\ 0 & 1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 1 & \frac{-3}{33} & \frac{22}{33} & 33 \end{array}\right]$$

$10\cdot F3+\widetilde{F1}$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{11} & \frac{-140}{3} & 330 \\ 0 & 1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 1 & \frac{-3}{33} & \frac{22}{33} & 33 \end{array}\right]$$

$-2\cdot F3+\widetilde{F2}$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{11} & \frac{-140}{3} & 330 \\ 0 & 1 & 0 & \frac{-2}{11} & \frac{-1}{3} & 66 \\ 0 & 0 & 1 & \frac{-3}{33} & \frac{22}{33} & 33 \end{array}\right]$$

$$A^{-1} = \begin{bmatrix} \frac{1}{11} & \frac{-140}{3} & 330 \\ \frac{-2}{11} & \frac{-1}{3} & 66 \\ \frac{-3}{33} & \frac{22}{33} & 33 \end{bmatrix}$$